AI Powered Intelligent Hospital Patient Flow Management System

BLOCK DIAGRAMS - ALL

Student	Hall Ticket No.
A. RAMU	23K91A6715
B. RUSHI	23K91A6720
B. RAJ KUMAR	23K91A6725
A. PRABHAS	23K91A6703

Under the guidance of [GUIDE NAME]

Department of Computer Science and Engineering (Data Science)

[COLLEGE NAME]

[ACADEMIC YEAR]

Diagram Inventory

The following diagrams were generated from the implemented SQLAlchemy entities, FastAPI routes, services, model inference functions and React role flows. They describe the checked-in code rather than a proposed future architecture.

System architecture

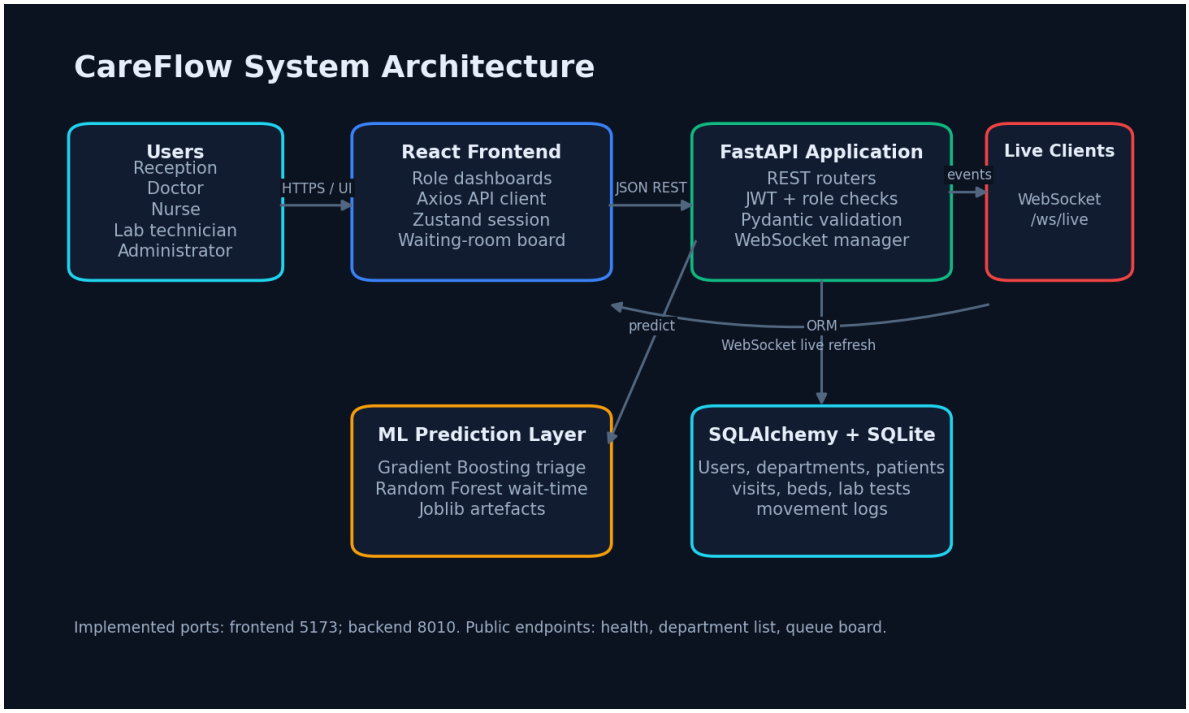


Figure 1. System architecture

The React client calls FastAPI REST routes, which coordinate JWT role checks, SQLAlchemy persistence and the two saved estimators. WebSocket events refresh live clients.

High-level patient workflow

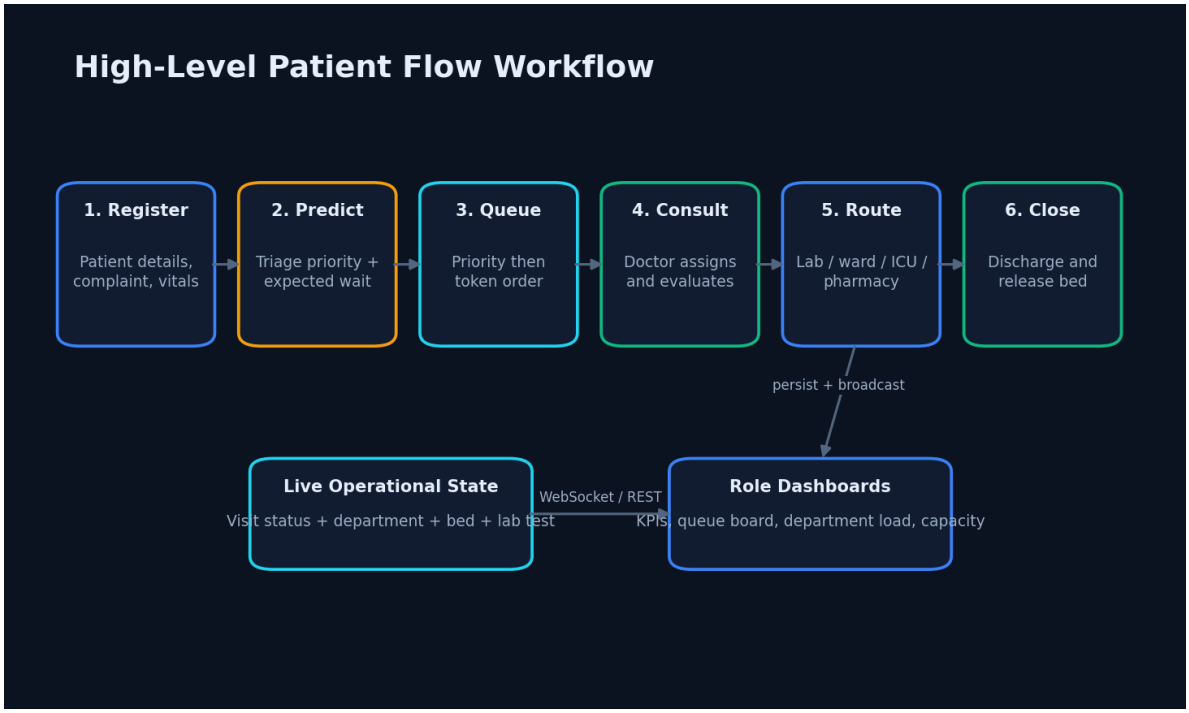


Figure 2. High-level patient workflow

The operational path runs from registration and prediction through priority queueing, consultation, routing and closure.

DFD Level 0

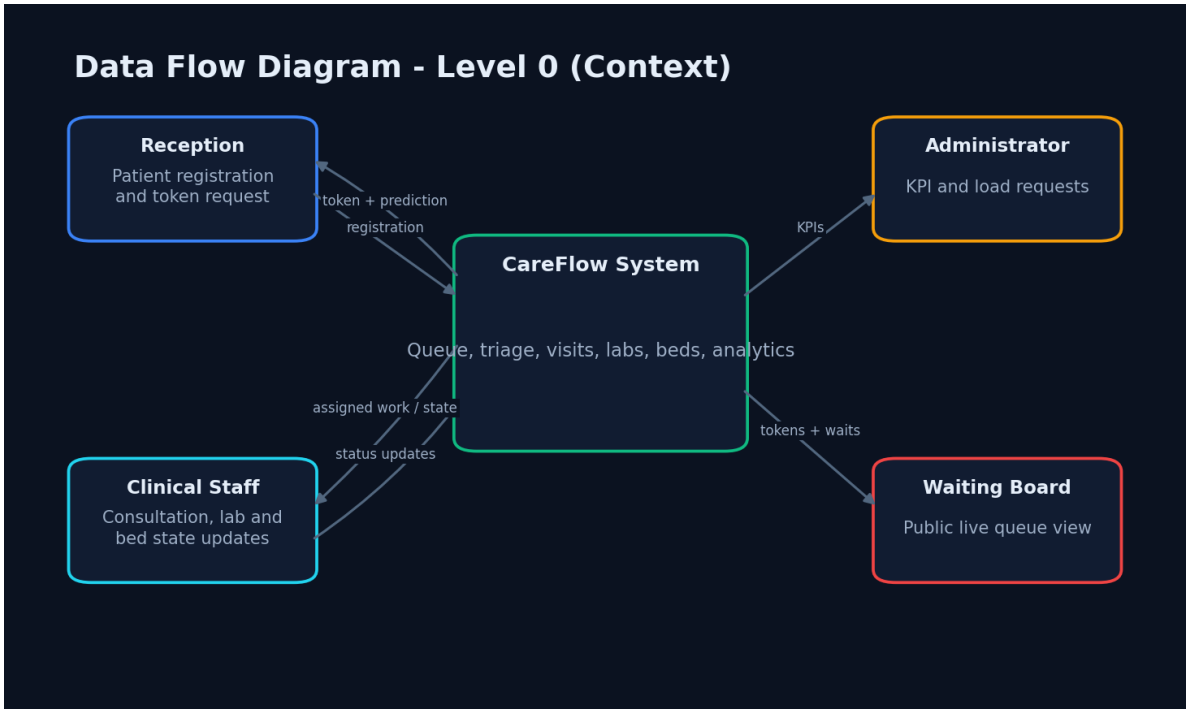


Figure 3. DFD Level 0

The context view shows reception, clinical staff, administration and the waiting board exchanging information with CareFlow.

DFD Level 1

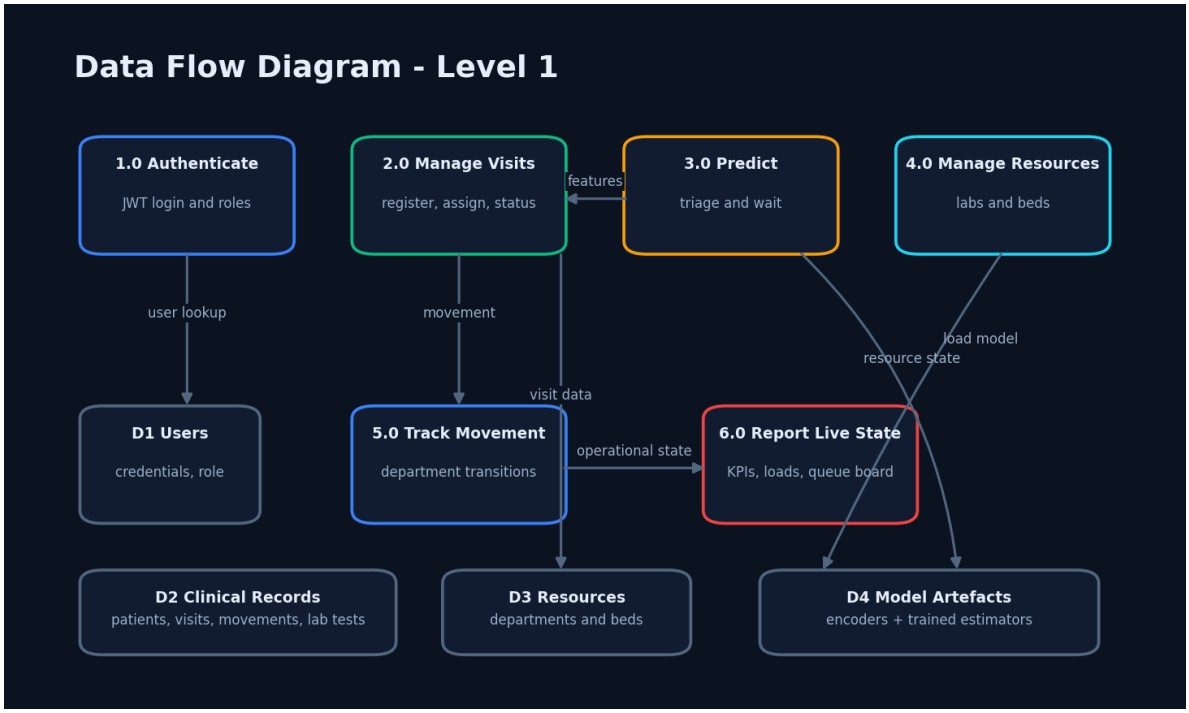


Figure 4. DFD Level 1

The system is decomposed into authentication, visit, prediction, resource, movement and reporting processes.

DFD Level 2

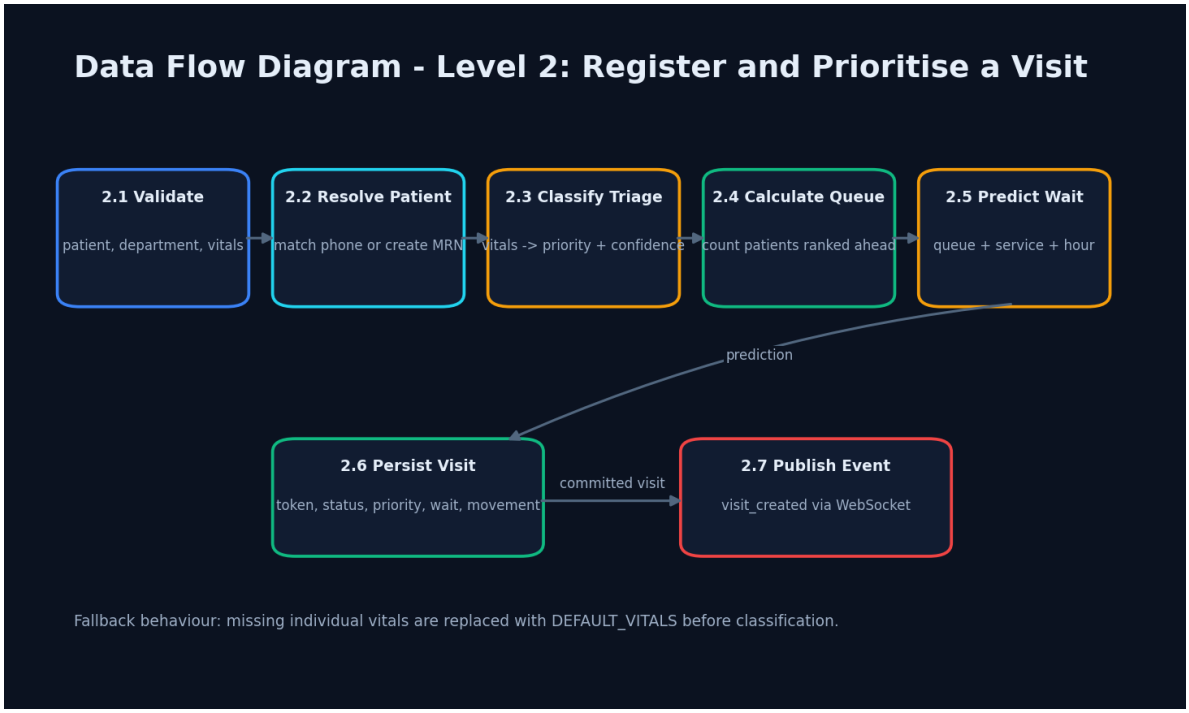


Figure 5. DFD Level 2

Visit creation validates inputs, resolves a patient, predicts priority and wait, persists the visit and publishes an event.

UML class diagram

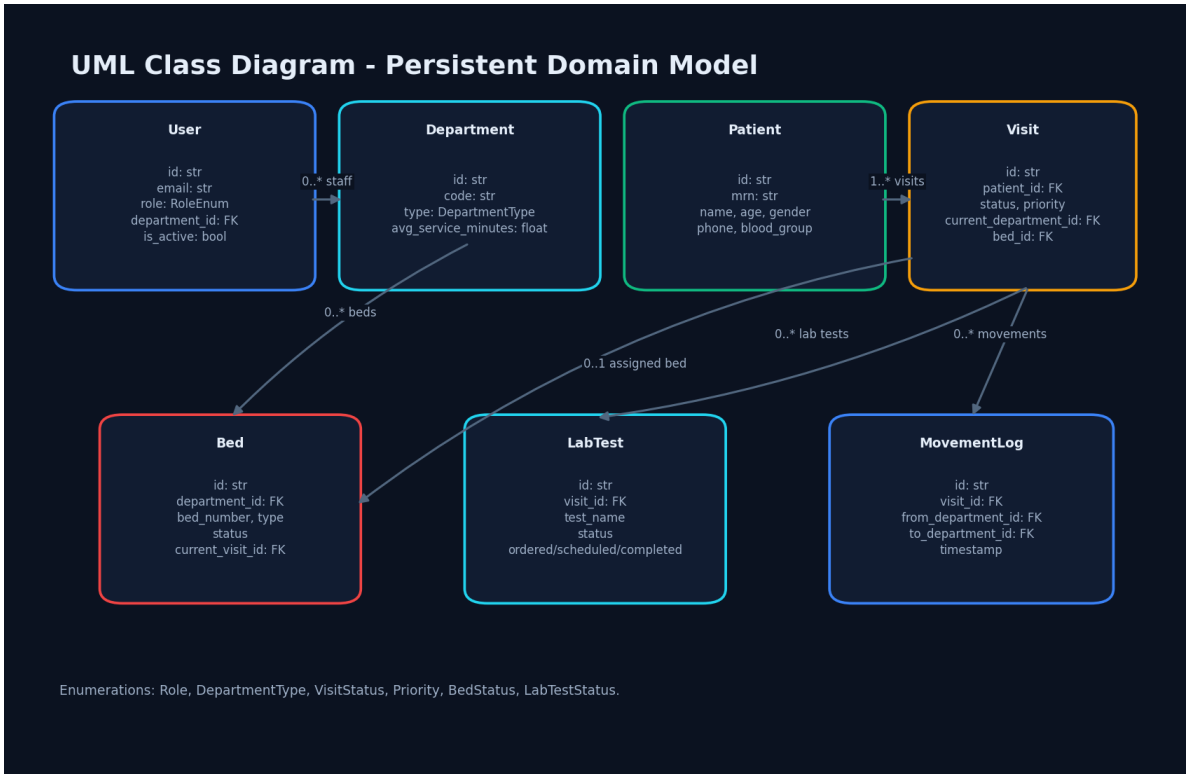


Figure 6. UML class diagram

The class view records the persistent model types and their operational relationships.

Registration sequence

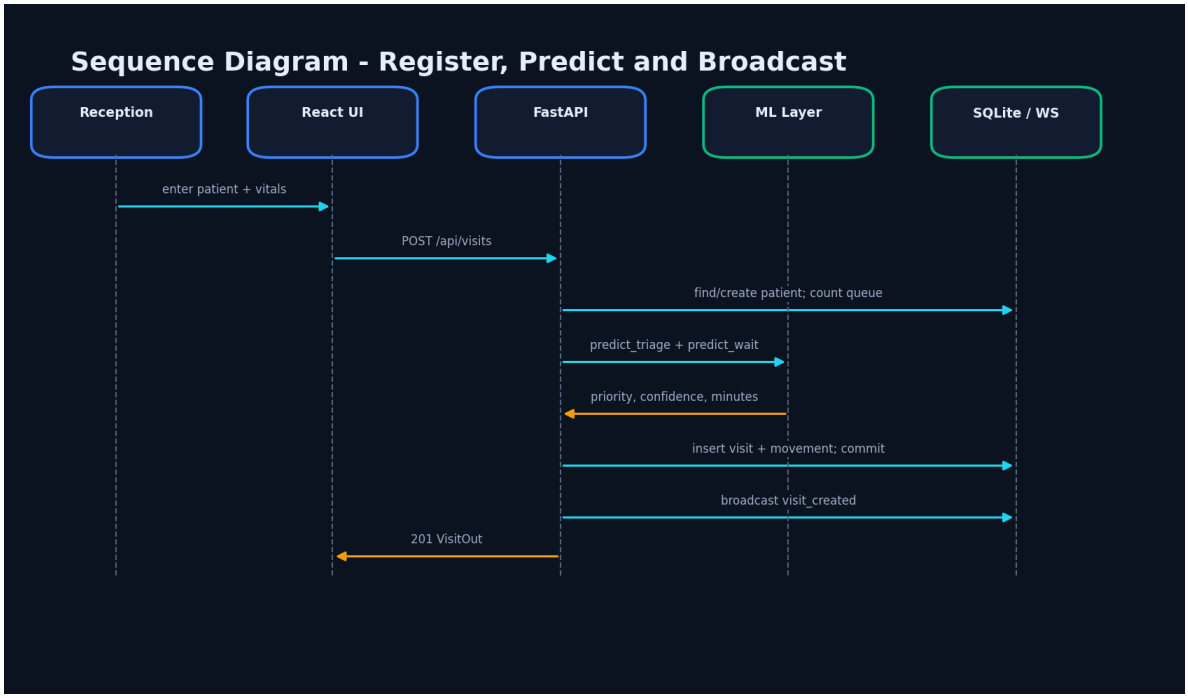


Figure 7. Registration sequence

Reception, frontend, API, ML layer, database and WebSocket participate in one registration transaction.

Use-case diagram

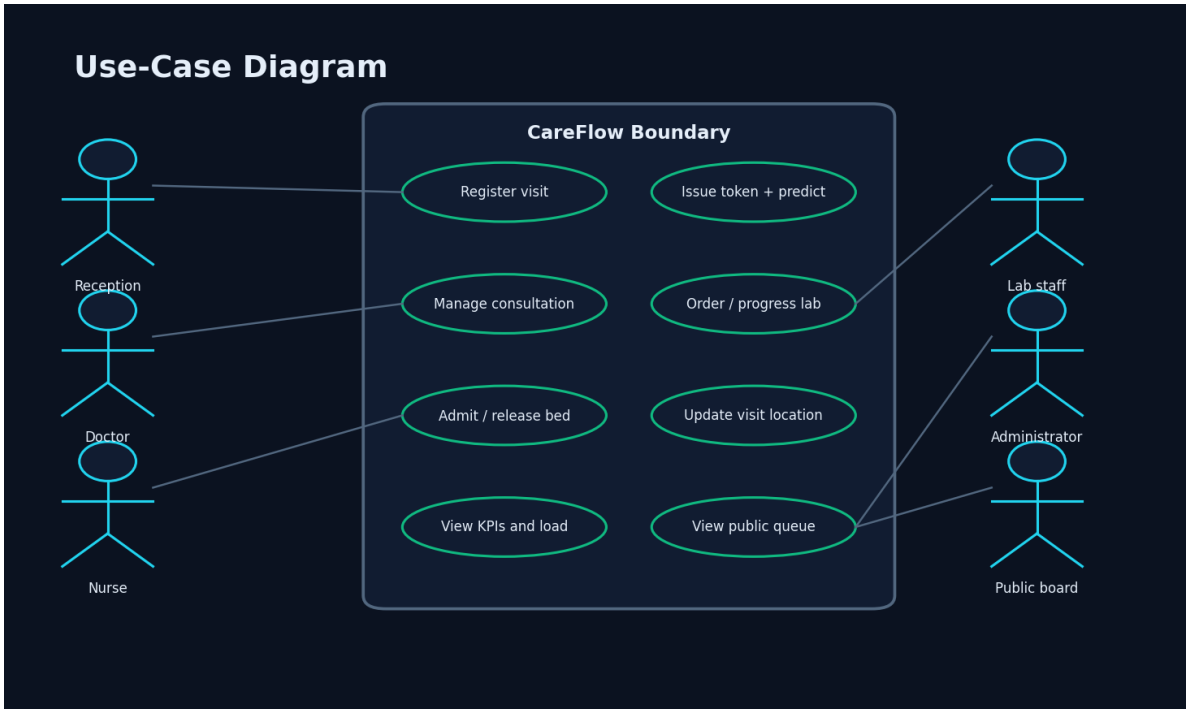


Figure 8. Use-case diagram

Each role is linked only to the functions implemented by its dashboard and API permissions.

Entity-relationship diagram

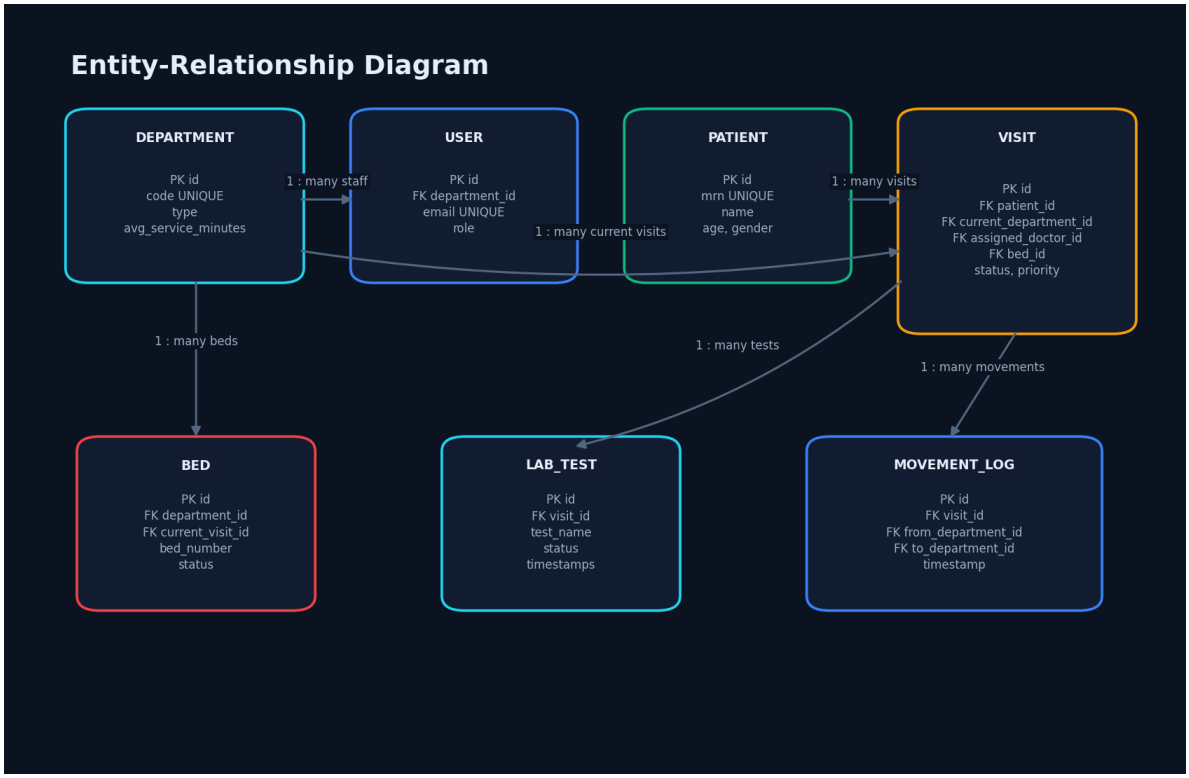


Figure 9. Entity-relationship diagram

Foreign-key paths connect departments, staff, patients, visits, beds, tests and movement logs.

Core algorithm flowchart

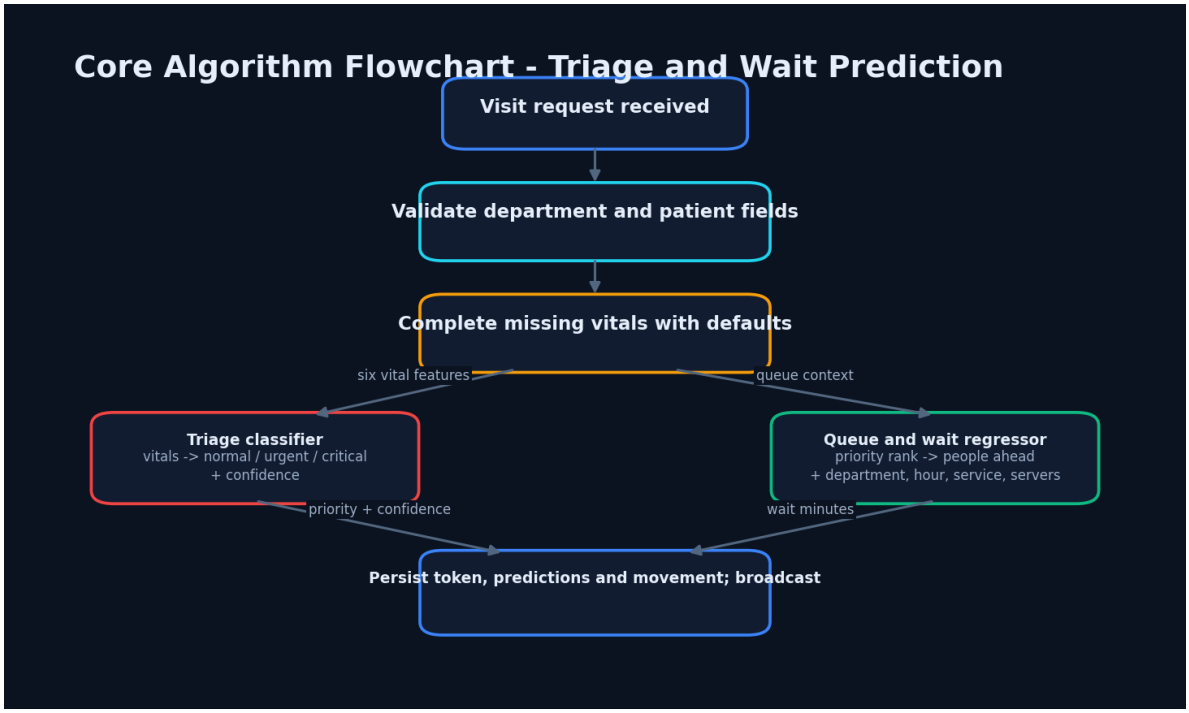


Figure 10. Core algorithm flowchart

The prediction flow combines vital-sign classification and queue-context regression before persistence.